

Suppose we want to train a Reinforcement Learning agent to play the game of [Tic-Tac-Toe](<https://en.wikipedia.org/wiki/Tic-tac-toe>), and need to construct an environment with states and actions. Assume our agent will simply choose actions based on the current state of the game, rather than trying to guess what the opponent will do next.

a. What should be the states and actions within the Tic-Tac-Toe Reinforcement Learning environment? Don't try to list them all, just describe how the rules of the game define what states and actions are possible. How does the current state of the game affect the actions you can take?

The Tic-Tac-Toe environment consists of a 3x3 grid where each player can place their symbol, either 'X' or 'O', on an empty cell of the grid. The game starts with an empty grid, and each player takes turns placing their symbol on a cell until one of the players wins or the game ends in a draw.

The state of the environment at any point in the game can be represented as a 3x3 matrix where each cell is either empty, contains an 'X' symbol, or an 'O' symbol. The state also needs to keep track of which player's turn it is. The possible actions that the agent can take depend on the current state of the game. The agent can choose any empty cell on the grid to place their symbol. If they choose a cell that is already occupied, their action is invalid and the agent must choose a different action.

The agent's ultimate goal is to win the game, but it can also aim to maximize its chances of winning by blocking the opponent's moves and creating opportunities for its own moves. The agent receives a positive reward for winning, a negative reward for losing, and zero reward for a draw.

b. Design a reward function for teaching a Reinforcement Learning agent to play optimally in the Tic-Tac-Toe environment. Your reward function should specify a reward value for each of the 3 possible ways that a game can end (win, loss, or draw) as well as a single reward value for actions that do not result in the end of the game (e.g., your starting move). Explain your choices.

One possible reward function for Tic-Tac-Toe would be:

1. Winning: +10 - The agent receives a positive reward of 10 when it wins the game, reflecting the ultimate goal of the agent in Tic-Tac-Toe.

2. Losing: -10 - The agent receives a negative reward of 10 when it loses the game, indicating that losing the game is highly undesirable.
3. Draw: 0 - The agent receives zero reward when the game ends in a draw, reflecting that the agent neither wins nor loses in this case.
4. Intermediate moves: +1 - The agent receives a small positive reward of 1 for each valid move it makes that does not result in the end of the game. This encourages the agent to make moves that increase its chances of winning, such as blocking the opponent's moves and creating opportunities for its own moves.

The reward function is designed to encourage the agent to prioritize winning the game, but also encourages the agent to make moves that increase its chances of winning, even if the immediate reward for a single move is low. The intermediate reward for valid moves also helps the agent to explore the state space and learn the optimal strategy.

c. Assume your agent gets to play first (and then alternates turns with the opponent). For actions that do not end the game, should the environment give rewards to the agent before or after the opponent plays? Why?

Yes; for actions that do not end the game, the environment should give rewards to the agent after the opponent plays. This is because the agent's action may result in a suboptimal state once the opponent has made their move, and the reward for the agent's action should reflect the final outcome of the game, not just the initial state after the agent's move.

By giving the reward after the opponent plays, the environment can take into account the opponent's move and determine whether the agent's previous move was a good one. This allows the environment to incentivize the agent to make moves that not only create opportunities for itself but also block the opponent's moves.

If the environment gives rewards to the agent before the opponent plays, it may incentivize the agent to make moves that are not optimal in the long run. For example, the agent may make a move that gives it an immediate reward but opens up a winning opportunity for the opponent in the next turn. By delaying the reward until after the opponent plays, the environment encourages the agent to consider the opponent's possible responses and make moves that lead to a more favorable outcome in the long run.